Evidence-Centric Forensic Knowledge Graphs: Ontology-Constrained Multimodal Extraction with Validity-Aware Evaluation

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Abstract—Forensic evidence is multimodal and siloed: textual sources (FIRs, court judgments, post-mortem, laboratory, deposition, and scene reports) sit apart from visual evidence (bloodstains, fingerprints, wounds, ballistic and tool marks). The cross-modal and cross-case relationships that drive investigation are therefore never made explicit. We present an ontology-guided, multimodal forensic knowledge-graph framework that extracts, type-validates, links, and reasons over evidence from both text and images: a forensic ontology (38 node types, 48 relationship types) constrains schema-bounded large language model (LLM) extraction over six document types and a hybrid computer-vision/LLM pipeline over six image types, populating one typed graph that supports semantic retrieval, grounded query generation, and conservative cross-case entity resolution. Our central methodological claim is that forensic AI must keep three notions distinct—schema compliance, extraction consistency, and external factual validity—and we build an evaluation that does. We separate circular synthetic-gold regression from validation against the documented facts of 28 real Indian criminal cases (gold from two independent annotators; inter-annotator Cohen's $\kappa = 0.95$ on roles, 1.0 on crime type). Over three runs, the system recovers persons, crime types, weapons, and dated events with strong, stable performance (F1 = 0.88, 0.91, 0.72, 0.88; standard deviation ≤ 0.01), with overall micro-/macro-F1 = 0.78/0.73 (bootstrap 95% CI $[0.74, 0.82]$). An ablation shows ontology guidance is decisive: it alone recovers weapons (F1 $0.00 \rightarrow 0.72$) and lifts crime-type ($0.75 \rightarrow 0.91$), location ($0.13 \rightarrow 0.27$), and macro-F1 ($0.53 \rightarrow 0.73$) over a naive LLM baseline. Image analysis is reliable only where ground truth is task-aligned (full-set bloodstain F1 = 1.0, n=61); misaligned tasks score near zero, which we report as evidence of evaluation honesty. The contribution is this validity-aware framework and its auditable reasoning—not a system that replaces forensic experts.

Index Terms—knowledge graphs, digital forensics, multimodal information extraction, large language models, entity resolution, retrieval-augmented reasoning, evaluation.

I. Introduction

A criminal case is, in essence, a web of relationships among people, places, times, actions, and physical traces. Yet the documentary and physical record of that case is fragmented across institutional boundaries: a police FIR, a forensic laboratory report, an autopsy, a sequence of witness statements, a scene-of-crime officer (SOCO) inventory, and a court judgment are typically authored by different agencies, in different formats, and stored in different systems. Visual evidence—photographs of bloodstains, fingerprints, wounds, fired projectiles, tool marks, and questioned documents—lives in yet another silo. Investigators and analysts must mentally reconstruct the connections that span these artefacts, and they must do so repeatedly, case after case, without the benefit of an explicit, queryable representation that ties the evidence together.

Two prevailing computational approaches each address only part of the problem. Flat relational or document databases store records efficiently but encode relationships implicitly through foreign keys or full-text search; multi-hop, cross-modal questions ("which cases share a weapon type, an investigating officer, and a cause of death?") become brittle joins or are simply unanswerable. Isolated LLM extraction, in which a language model reads a document and emits entities, produces locally plausible structure but no durable, validated, interconnected representation: outputs are unconstrained by a shared schema, are not linked across documents or modalities, and are prone to hallucinated entities and relations that no downstream check can catch.

What forensic reasoning needs is a representation that is (i) graph structured, so that relationships are first-class and traversable; (ii) ontology constrained, so that every extracted entity and relation is type-checked against a domain schema before it enters the store; (iii) multimodal, so that textual and visual evidence populate one shared graph; and (iv) evidence-centric and auditable, so that every reasoning output can be traced back to the specific nodes and edges that support or contradict it.

This paper presents such a framework. We design a forensic ontology and use it to constrain LLM-based extraction from six document types and a hybrid computer-vision/LLM pipeline for six image types, fusing both into a single typed property graph. Over that graph we define a reasoning layer for semantic retrieval, grounded natural-language querying, hypothesis support, conservative cross-

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case entity resolution, and temporal reconstruction. The contribution is this ontology-guided, multimodal framework together with its evaluation methodology—not a software product; a prototype implementation exists only to produce the measurements reported here.

A central argument runs through the paper. Much LLM-forensics work reports extraction quality against synthetic or model-generated gold, which risks circular validation: a model is scored against labels produced by the same class of model. We argue that forensic AI must keep three notions distinct: schema compliance (does the output satisfy the ontology?), extraction consistency (does the system reproduce a reference, possibly self-generated, label set?), and external factual validity (does the output match facts established independently of any model?). Conflating them inflates apparent performance. The framework is therefore built so that schema compliance is an invariant, consistency is measured only as a regression signal, and validity is measured against the documented facts of real cases—and, for vision, only where a dataset's ground truth actually operationalises the forensic task.

Contributions: We make the following contributions:

1) A validity-aware evaluation methodology for forensic LLM/KG systems—the paper's primary methodological contribution: the schema-compliance / consistency / validity distinction above, instantiated as (i) a documented-fact, two-annotator protocol (Cohen's $\kappa = 0.95/1.0$, section VII-C) that breaks circular evaluation, and (ii) a ground-truth-alignment analysis that explains why some multimodal forensic benchmarks cannot validate the task they appear to. The methodology is not merely a framing: applying it surfaced two concrete validity errors in our own pipeline—a label leak that inflated fingerprint accuracy from a true ≈ 0 to a false 100%, and a metric that scored recovered dates from the wrong field (F1 = 0.09 vs. the correct 0.88)—both diagnosed, corrected, and reported (sections VI and VII).
2) A domain-specific forensic ontology of 38 node types and 48 relationship types spanning legal, medical, evidentiary, temporal, scene, and visual-forensic entities, with type-checked graph construction as an invariant (section IV-A, table I, eqs. (1) and (2)).
3) Ontology-constrained multimodal extraction: a schema-registry text pipeline over six document types and a hybrid CV + LLM image pipeline over six image types, fused into one typed graph (sections IV-B to IV-D).
4) A conservative cross-case entity-resolution method with a non-destructive guarantee: candidate nodes are never merged, only linked by SAME_AS, so provenance is never collapsed and a wrong link stays recoverable—the failure mode that matters most in forensics. It pairs embedding-blocked candidate generation with name/role/context verification and, on a labelled 9,973-pair benchmark, attains precision 0.96 at the deliberately conservative recall the setting demands (sections IV-F and VII-E).
5) A graph-centred reasoning layer—semantic retrieval, grounded NL-to-graph querying, evidence-chain hypothesis support, and a temporal-reconstruction module over recovered dated events (TimeEvent F1 = 0.88) (sections IV-E and IV-G).
6) Honest negative results presented as scientific evidence: several image tasks fail under misaligned ground truth, which delimits the framework's true operating envelope and guards against inflated claims (sections VII and VIII).

Research questions: We organise the evaluation around four questions. RQ1: does ontology guidance improve forensic entity extraction over a naive LLM JSON baseline (section VII-G)? RQ2: does documented-fact validation reveal performance different from circular synthetic regression (sections VI-B and VII)? RQ3: which forensic image modalities have task-aligned ground truth suitable for evaluating multimodal extraction (section VII)? RQ4: can conservative, non-destructive SAME_AS linking achieve high-precision cross-case links while preserving provenance (section IV-F)?

We deliberately frame the system as decision and hypothesis support. It does not, and is not intended to, render forensic conclusions, and we make no claim about legal admissibility.

II. Related Work

A. Knowledge graphs for forensic and legal reasoning

Knowledge graphs (KGs) provide a relational substrate in which entities and their typed connections are explicit and queryable [1], [2]. Property-graph data models and their query languages [3] support the multi-hop traversal and pattern matching that are awkward in flat schemas, and systems such as Neo4j [4] implement them. In the legal and forensic domains, NLP and structured-representation methods have been applied to statutes, cases, and entities [5], [6], but most efforts focus on a single modality (legal text) and do not integrate physical or visual forensic evidence. Our work differs in unifying legal, medical, procedural, and visual forensic evidence within one ontology-constrained graph, and—more distinctively—in pairing that graph with an evaluation protocol that separates extraction consistency from external factual validity.

B. LLM-based information extraction

Transformer language models [7], [8] and large generative models [9], [10] have substantially advanced information extraction, moving from task-specific NER [11] and open IE [12] toward prompt-driven, schema-constrained generation. LLMs can emit structured outputs directly, but without an external schema and validation step their outputs drift and hallucinate. We constrain generation with per-document-type JSON schemas and validate every candidate against the ontology before insertion, combining the flexibility of LLM extraction with the guarantees of a typed store.

C. Multimodal forensic analysis

Forensic image analysis has historically been task-specific: bloodstain pattern analysis [13], [14], firearm and tool-mark comparison [15], fingerprint and document examination. Vision-language models [16] enable general image interpretation, but domain forensic tasks demand measurable, physics- or procedure-grounded outputs. We pair classical CV feature extraction with LLM vision interpretation under a strategy pattern, and—importantly—evaluate each modality against the most appropriate available ground truth, reporting where that ground truth is misaligned with the task.

D. Graph retrieval and semantic search

Retrieval-augmented generation grounds LLM outputs in retrieved context [17], and recent work extends retrieval to graph structure [18]. Translating natural language into graph query languages (e.g. Cypher) is an active area [18], [19]. We embed every graph node and use vector k-nearest-neighbour search both for direct semantic retrieval and to ground a natural-language-to-Cypher generator with the actual property values and relationship patterns present in the graph, reducing schema-mismatch errors.

E. Entity resolution in knowledge graphs

Record linkage and entity resolution have a long history [20]–[22]: the task is to decide when two records refer to the same real-world entity. In forensic settings a false merge is dangerous—it can fabricate a connection between unrelated cases or people. We therefore adopt a conservative, blocking-then-verify design and, critically, make resolution non-destructive: candidate duplicates are linked, never merged, preserving the provenance of every source mention.

F. Evaluation challenges for forensic AI

National reviews have stressed that forensic methods require demonstrated scientific validity and known error rates [23], [24], and the forensic-science literature has long documented both the absence of validated error rates for many feature-comparison disciplines [25] and the susceptibility of expert judgement to contextual bias [26]. These concerns motivate our insistence on task-aligned ground truth and on keeping consistency and validity separate. For AI systems, a recurring pitfall is circular evaluation: when "gold" annotations are produced by the same class of model being evaluated, high scores measure self-consistency rather than correctness. We address this by validating extraction against documented facts of real cases curated from authoritative public sources, and by being explicit about where image ground truth does not encode the attribute the task requires.

III. Problem Formulation

We formalise the framework as a pipeline from heterogeneous evidence to a typed, embedded, queryable case graph.

Inputs: Let $D = \{d_i\}$ be a corpus of textual forensic documents, each with a source type $s_i \in S$, where S comprises the six document types (FIR, court judgment, post-mortem, laboratory report, witness deposition, SOCO report). Let $I = \{x_j\}$ be a set of image evidence items, each with a modality $m_j \in M$, where M comprises the six image types (bloodstain, fingerprint, wound, ballistics, document, tool mark).

Ontology: A forensic ontology $O = (T_N, T_R, C)$ specifies node types T_N ($|T_N| = 38$), relationship types T_R ($|T_R| = 48$), and constraints C. Each relationship type $r \in T_R$ carries a domain/range signature $\mathrm{dom}(r), \mathrm{ran}(r) \subseteq T_N$; C additionally encodes required properties per node type and controlled enumerations for selected attributes.

Knowledge graph: The target is a property graph $G = (V, E, X)$ with a typing function $\tau : V \to T_N$, a set of typed edges $E \subseteq \{(u, r, v) \mid u, v \in V,\ r \in T_R,\ \tau(u) \in \mathrm{dom}(r),\ \tau(v) \in \mathrm{ran}(r)\}$, and an embedding map $X : V \to \mathbb{R}^d$ ($d = 1536$) defined as $X = \phi \circ \sigma$, where σ serialises a node's labels and properties to text and ϕ is a pretrained text-embedding function.

Validity: Write $\mathrm{req}(t)$ for the required properties of type t. A node v is valid and an edge (u, r, w) is admissible iff

$$\mathrm{valid}(v) \equiv \mathrm{req}(\tau(v)) \subseteq \mathrm{keys}(v) \ \wedge\ \mathrm{enums}(v), \quad (1)$$

$$\mathrm{adm}(u, r, w) \equiv \tau(u) \in \mathrm{dom}(r) \ \wedge\ \tau(w) \in \mathrm{ran}(r), \quad (2)$$

where $\mathrm{enums}(v)$ holds iff every enumerated attribute of v takes an allowed value. Only valid nodes and admissible edges enter G; this invariant is preserved by construction (section IV-A).

Extraction: Text extraction is a function $f_{\text{text}} : (d, s) \mapsto (V_d, E_d)$ that respects O; we decompose it as $f_{\text{text}} = f_{\text{rel}} \circ f_{\text{ent}}$ (entities first, then relations over those entities). Image analysis is $f_{\text{img}} : (x, m) \mapsto (V_x, E_x, z_x)$, where z_x is a vector of classical CV features that conditions and is fused with the LLM interpretation.

Fusion: A fusion operator Φ validates and merges modality outputs into a case graph, $\Phi\big((V_d, E_d), (V_x, E_x)\big) \mapsto G_c$, discarding any node or edge that violates O and inserting cross-modal links where a textual entity and an image-derived finding co-refer.

Resolution: A resolution operator R maps a candidate node pair to a link decision, $R(v_i, v_j) \mapsto \{\mathrm{SAME\_AS}, \varnothing\}$, used to connect co-referent entities across cases without merging them.

Reasoning: Finally, a reasoning function answers a natural-language question q by retrieving context, generating and executing a graph query, and interpreting the result; a hypothesis function maps a case c to a primary hypothesis with explicit supporting and contradicting evidence sets $(\mathcal{S}_c, \mathcal{C}_c) \subseteq V$.

IV. Methodology

Figure 1 shows the end-to-end architecture: heterogeneous evidence is processed by modality-specific extractors, validated against the ontology, materialised in a property graph with per-node embeddings, and served to a graph-centred reasoning layer that also writes derived artefacts (e.g. hypotheses) back into the graph.

A. Ontology-Guided Forensic Knowledge Graph

The ontology is the backbone of the framework. It defines six semantic groups of node types—core case, legal/judicial, medical/laboratory, scene/procedural, forensic-image, and reasoning entities—together with the typed relationships that connect them (fig. 2, table I). Core case entities (Case, Person, Location, Evidence, Weapon, TimeEvent, …) anchor every case. Legal entities (LegalSection, Verdict, CourtOrder) capture judicial structure; medical and laboratory entities (InjuryPattern, CauseOfDeath, ToxicologyResult, DNAProfile, …) capture forensic-medical findings; scene/procedural entities (CrimeScene, PhysicalEvidence, ChainOfCustody, Statement) capture investigative process; and image entities (BloodstainPattern, FingerprintPattern, WoundPattern, BallisticsPattern, ToolMarkPattern, …) capture visual analysis outputs. Reasoning entities (Hypothesis, ImpactMechanismNode, Experiment) host derived knowledge.

Ontology validation is what separates this framework from unconstrained LLM extraction. Graph construction is a left fold of a validated-insertion operator $\oplus$ over the per-document and per-image subgraphs:

$$G = \Big(\bigoplus_i (V_{d_i}, E_{d_i})\Big) \oplus \Big(\bigoplus_j (V_{x_j}, E_{x_j})\Big), \qquad (3)$$

where $\oplus$ inserts only the valid nodes $\{v : \mathrm{valid}(v)\}$ and admissible edges $\{e : \mathrm{adm}(e)\}$ of eqs. (1) and (2), coercing each admitted node's properties to their declared types and merging nodes that share a type's unique key. The invariant "G contains only valid nodes and admissible edges" therefore holds after every insertion. Two benefits follow. First, reliability: malformed or hallucinated structures are rejected before they reach the store, so reasoning operates over a consistent representation. Second, query safety: because the schema is fixed, generated queries can be grounded in real labels and edge patterns (section IV-E) and a static guard can reject destructive operations.

B. Multi-Document Text Extraction

Text extraction maps six structurally different document types onto the same graph target through a schema registry (table II). Each source type s is associated with an entity schema, a relationship schema, and a specialised system prompt. Extraction proceeds in two stages (algorithm 1): first, entities are extracted under the entity schema with the document-type prompt; second, relationships are extracted conditioned on the already-identified entities, so that the model links real referents rather than inventing endpoints. Both stages use strict structured output constrained to the schema, and every relationship is required to carry the text span that supports it, enabling provenance and auditability.

Algorithm 1 Two-stage, schema-constrained text extraction

Require: document d, source type s, ontology O, registry $\mathcal{R}$
Ensure: validated subgraph (V_d, E_d)
1: $(\Pi_{\mathrm{ent}}, \Sigma_{\mathrm{ent}}) \leftarrow \mathcal{R}[s]$ ▷ prompt + entity schema
2: $\tilde{V} \leftarrow \mathrm{LLM}(\Pi_{\mathrm{ent}}, d)$ constrained to Σ_{ent}
3: $V_d \leftarrow \{\, \mathrm{Coerce}(v, O) : v \in \tilde{V},\ \mathrm{Valid}(v, O) \,\}$
4: $(\Pi_{\mathrm{rel}}, \Sigma_{\mathrm{rel}}) \leftarrow \mathcal{R}[s]$
5: $\tilde{E} \leftarrow \mathrm{LLM}(\Pi_{\mathrm{rel}}, d, \mathrm{Summary}(V_d))$ constrained to Σ_{rel}
6: $E_d \leftarrow \{\, e \in \tilde{E} : \mathrm{TypeCheck}(e, O) \ \wedge\ \mathrm{ends}(e) \subseteq V_d \,\}$
7: $\mathrm{PrefixIds}(V_d, E_d, \mathit{docid})$ ▷ namespace per document
8: return (V_d, E_d)

The two-stage decomposition matters: relationship extraction is the harder, more error-prone step, and grounding it in a fixed entity set both reduces the combinatorial space and prevents the model from introducing entities through the back door of a relationship. Document-type prompts inject domain knowledge (e.g. that a post-mortem's deceased subject is a Person with a victim role, that a CauseOfDeath should be linked to that person via CAUSE_OF_DEATH_OF), while the common graph target guarantees that heterogeneous documents become interoperable subgraphs.

C. Hybrid Forensic Image Analysis

Each image modality is handled by a strategy implementing a common four-stage map $f_{\mathrm{img}} = \mathrm{build} \circ \mathrm{combine} \circ \langle \mathrm{cv}, \mathrm{vision} \rangle$. The cv stage runs classical computer vision (e.g. CLAHE enhancement, Otsu segmentation, and contour-based geometric and spatial feature extraction for bloodstain images), producing the feature vector z_x. The vision stage queries an LLM vision model with a domain prompt; combine synthesises z_x, the vision output, and any experiment metadata into a structured assessment; and build maps that assessment to ontology-typed entities and relationships. The CV stage contributes quantitative per-stain geometry and spatial statistics to the graph; an ablation (section VII) finds that, on our impact-spatter sample, the LLM vision step alone already detects the pattern, so the hybrid's CV features add graph structure rather than classification accuracy on that coarse label.

Table III summarises the six modalities, the CV features each computes, the semantic output produced, and the ground truth against which it is evaluated. We stress an honest point that the evaluation (section VII) makes quantitative: only some image types have ground truth that ac-

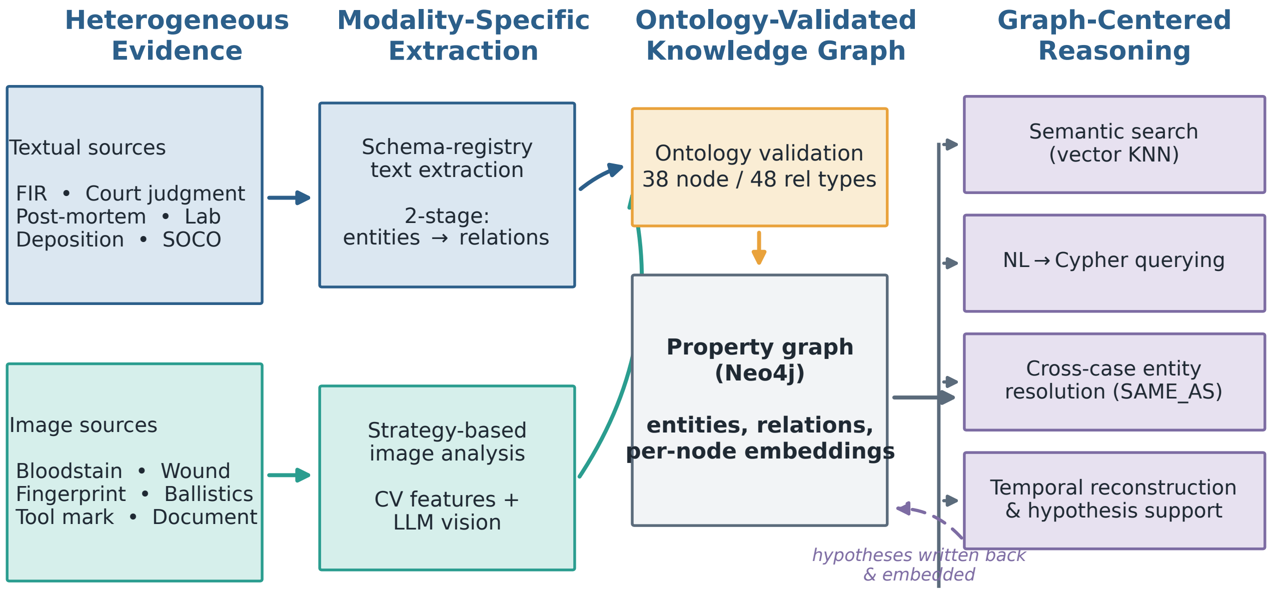

Fig. 1. End-to-end architecture. Textual and visual evidence are extracted by modality-specific components, type-checked against the forensic ontology, materialised in an embedded property graph, and consumed by semantic search, NL-to-Cypher querying, cross-case entity resolution, temporal reconstruction, and hypothesis support. Derived artefacts are written back and embedded.

tually encodes the attribute the task requires. Bloodstain analysis is evaluated against a physics-grounded impact-spatter dataset; fingerprint labels in the public dataset encode hand and finger position, not pattern class, so they cannot validate pattern recognition. We retain all six modalities in the framework but report their evaluation faithfully.

D. Multimodal Fusion

Fusion (fig. 3) inserts both branches' outputs into a single case graph. Text and image subgraphs share the case namespace, and the fusion operator adds cross-reference edges where a textual entity and an image-derived finding co-refer (for example, a Weapon mentioned in an FIR and a BallisticsPattern extracted from a fired-cartridge image, linked via FIRED_FROM; or a WoundPattern from a wound image linked to the weapon that a post-mortem attributes it to via WOUND_CAUSED_BY). Because the two branches are independent, fusion degrades gracefully: a text-only case still yields a valid case graph, as does an image-only case. This robustness is a direct consequence of the shared ontology—each branch produces typed, self-contained structure that the graph can absorb in isolation or together.

E. Semantic Retrieval and NL-to-Graph Querying

Each node is serialised by σ and embedded by ϕ, giving $X(v) = \phi(\sigma(v)) \in \mathbb{R}^{1536}$; a single cosine vector index over a shared label covers all node types. Two capabilities follow. Semantic search returns, for a query q, the k nearest nodes

$$\mathcal{N}_k(q) = \arg\operatorname*{top\text{-}k}_{v\in V} \cos\big(\phi(q), X(v)\big), \quad \cos(a,b) = \frac{\langle a,b\rangle}{\|a\|\,\|b\|}. \tag{4}$$

Grounded NL-to-graph querying conditions generation on real graph content. From $\mathcal{N}_k(q)$ we collect the retrieved nodes' stored property values and the set of realised schema edges

$$\mathcal{P}(\mathcal{N}_k) = \big\{(\tau(u), r, \tau(w)) : (u,r,w) \in E, \quad \{u,w\} \cap \mathcal{N}_k \neq \varnothing\big\}, \tag{5}$$

and pass $(q, \text{props}, \mathcal{P})$ to the generator, so it filters on values that exist and traverses edges that exist—e.g. (:InjuryPattern)-[:LED_TO_DEATH]->(:CauseOfDeath)—rather than hypothesising schema. A bounded loop repairs an erroneous query by returning the database error to the generator (at most a few attempts), and a static guard rejects destructive operations. These are means to grounded, trustworthy querying rather than contributions in themselves.

F. Cross-Case Entity Resolution

Each document is extracted into its own namespace, so the same real-world entity—a recurring judge, a shared location, a weapon type—appears as distinct nodes across cases. Resolution links these without merging them (fig. 4, algorithm 2).

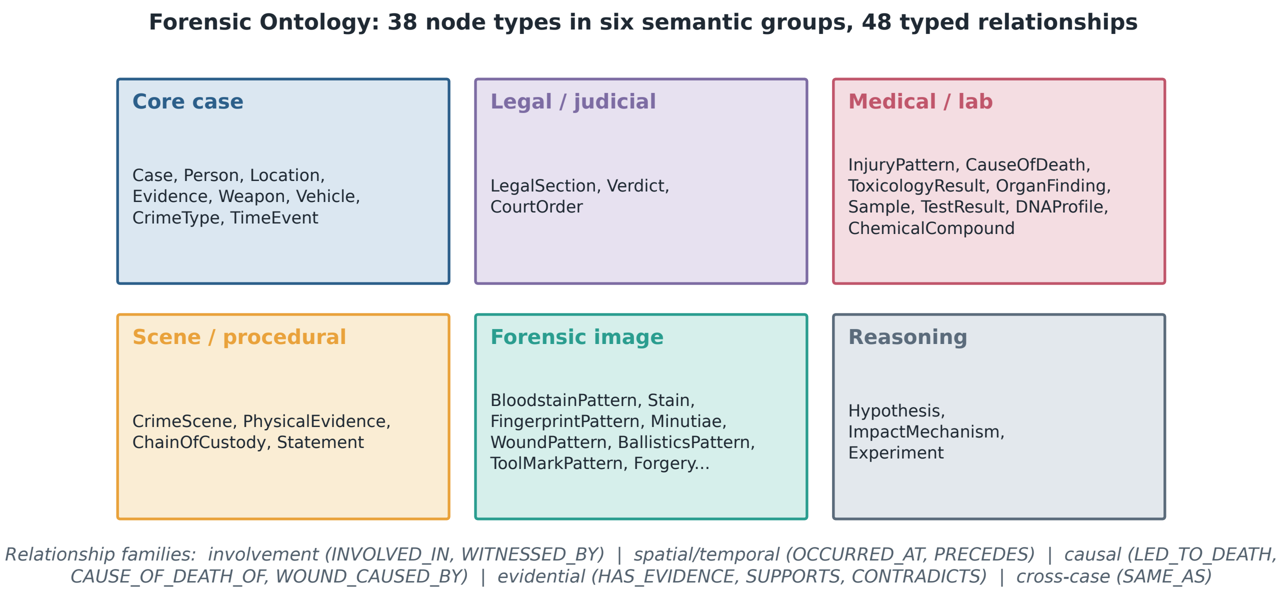

Fig. 2. The forensic ontology groups 38 node types into six semantic families and connects them with 48 typed relationships organised into involvement, spatial/temporal, causal, evidential, image-derived, and cross-case families.

TABLE I
The forensic ontology organises 38 node types into six semantic groups and connects them with 48 typed relationships. Required properties and controlled enumerations (e.g. PatternType, InjuryType, FingerprintClass) are enforced at insertion time.

Semantic group	# types	Representative node types
Core case	8	Case, Person, Location, Evidence, Weapon, Vehicle, CrimeType, TimeEvent
Legal / judicial	3	LegalSection, Verdict, CourtOrder
Medical / lab	8	InjuryPattern, CauseOfDeath, ToxicologyResult, OrganFinding, Sample, TestResult, DNAProfile, ChemicalCompound
Scene / procedural	4	CrimeScene, PhysicalEvidence, ChainOfCustody, Statement
Forensic image	12	BloodstainPattern, Stain, FingerprintPattern, MinutiaePoint, RidgeDetail, WoundPattern, TissueAnalysis, BallisticsPattern, HandwritingFeature, InkAnalysis, ForgeryIndicator, ToolMarkPattern
Reasoning	3	Hypothesis, ImpactMechanismNode, Experiment
Relationship families: involvement (INVOLVED_IN, WITNESSED_BY, PRESIDED_BY); spatial/temporal (OCCURRED_AT, HAPPENED_ON, PRECEDES, FOLLOWS); causal/medical (LED_TO_DEATH, CAUSE_OF_DEATH_OF, WOUND_CAUSED_BY, INJURY_CAUSED_BY); evidential (HAS_EVIDENCE, SUPPORTS, CONTRADICTS, CORROBORATES); image-derived (HAS_PATTERN, HAS_MINUTIAE, FIRED_FROM); and cross-case (SAME_AS).		

Blocking. Rather than comparing all $\Theta(n^2)$ cross-namespace pairs, we restrict comparison to the candidate set

$$\mathcal{B} = \big\{(i,j) : \mathrm{ns}(i) \neq \mathrm{ns}(j),\ \cos(X_i, X_j) \geq \theta_{\text{block}}\big\}, \quad (6)$$

using the node embeddings of eq. (4) to keep only semantically near pairs.

Verification. Let ν normalise a name (lowercasing; stripping honorifics, role words, and non-alphanumerics). For $(i,j) \in \mathcal{B}$ let $\sigma_{ij} \in [0,1]$ be the normalised-name similarity (a sequence-overlap ratio with a subset bonus for multi-token containment), $\rho_{ij} = \mathbf{1}[r_i = r_j]$ role agreement, and

$$\omega_{ij} = \frac{|\mathrm{ctx}_i \cap \mathrm{ctx}_j|}{|\mathrm{ctx}_i \cup \mathrm{ctx}_j|} \quad (7)$$

the Jaccard overlap of the nodes' 2-hop context bags (neighbour names, excluding Case and SAME_AS). The pair is accepted iff

$$\sigma_{ij} \geq \theta_{\text{name}} \ \wedge\ \big(\mathrm{unamb}(i,j) \vee \omega_{ij} > 0 \vee \mathrm{distinct}(r_i, r_j)\big), \quad (8)$$

where unamb holds when the shorter name is multi-token with no conflicting initials and distinct holds when both share a recurring distinctive role (judge, investigating officer). Weak signals—single-token or common names,

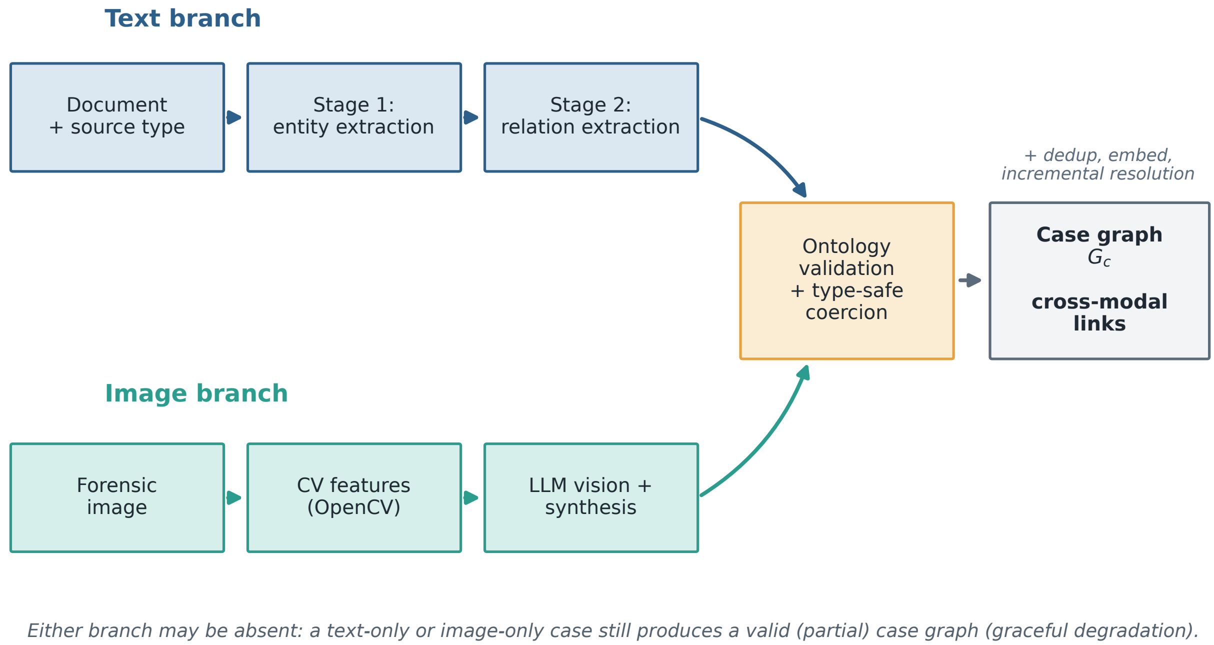

Fig. 3. Multimodal extraction workflow. The text branch performs two-stage entity/relation extraction; the image branch fuses CV features with LLM vision. Both pass through ontology validation before insertion into the case graph G_c; either branch may be absent (graceful degradation).

TABLE II
The schema registry binds each document type to an entity schema, a relation schema, and a prompt; all share the common graph target. Key entity types and a representative relation are shown.

Source type	Key entity types	Example relation
FIR	Case, Person, Location, Weapon, Evidence, TimeEvent	INVOLVED_IN
Court judgment	Person, LegalSection, Verdict, CourtOrder	CHARGED_UNDER
Post-mortem	InjuryPattern, CauseOfDeath, ToxicologyResult	LED_TO_DEATH
Lab report	Sample, TestResult, DNAProfile, Compound	MATCHES_DNA
Deposition	Statement, Person, TimeEvent	STATED_BY
SOCO	CrimeScene, PhysicalEvidence, ChainOfCustody	COLLECTED_BY

conflicting initials—therefore cannot link without corroborating context or role.

Non-destructive linking. An accepted pair receives a transparent confidence

$$c_{ij} = 0.6\,\sigma_{ij} + 0.25\min(\omega_{ij}, 1) + 0.15\,\rho_{ij} \tag{9}$$

and is joined by an edge $i \xrightarrow{\mathrm{SAME\_AS}\{c_{ij}\}} j$; connected

TABLE III
Forensic image modalities: classical CV features, semantic output, and the ground truth used for evaluation. Alignment varies by modality and is quantified in table VIII.

Modality	CV features	Semantic output	Eval. target
Bloodstain	CLAHE, Otsu; geometry, spatial stats	pattern + mechanism	impact-spatter (aligned)
Fingerprint	ridge, minutiae cues	pattern class	hand, finger (misaligned)
Wound	region, colour cues	wound type + mechanism	present / absent (partial)
Ballistics	striation cues	caliber, rifling	caliber (proxy)
Document	layout, ink cues	forgery indicators	genuine / forged (hard)
Tool marks	striation, impression cues	tool class	tool type (misaligned)

components (union–find) yield cross-case clusters. Because $|V|$ is unchanged, provenance is preserved exactly: no node is merged or deleted. This reflects a forensic priority—a false merge fabricates a connection between unrelated cases and is far more damaging than a missed link.

G. Temporal Reasoning and Hypothesis Support

The prototype includes a temporal reasoning module over TimeEvent nodes attached to a case. As section VII reports, the pipeline recovers the documented dates well (TimeEvent F1 = 0.88, date coverage 0.95), so this module operates on reliable dated events. The

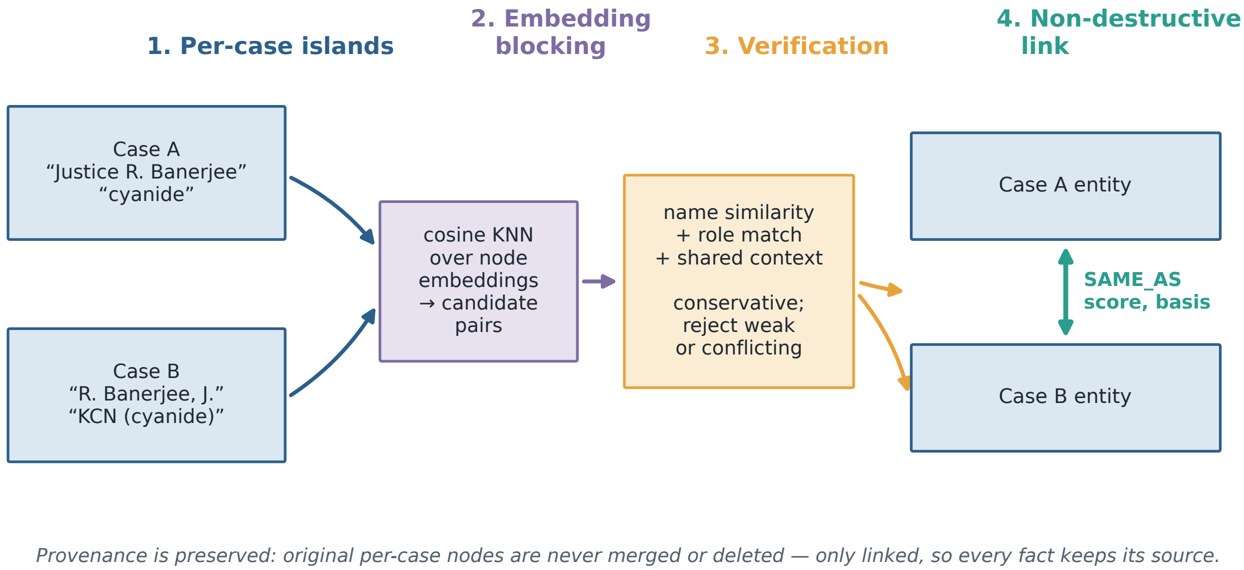

Fig. 4. Cross-case entity resolution. Per-case islands are compared via embedding-blocked candidate generation, verified by name/role/context, and linked with non-destructive SAME_AS edges that preserve provenance.

Algorithm 2 Incremental, non-destructive cross-case resolution

Require: new namespace p; resolvable labels L; thresholds $\theta_{\text{block}}, \theta_{\text{name}}$
Ensure: SAME_AS links connecting namespace p to the rest of G

$\Lambda \leftarrow \varnothing$
for each label $\ell \in L$ **do**
  $N \leftarrow$ ℓ-nodes with embedding X, role r, 2-hop context ctx
  normalise names via ν; discard generic placeholders
  for each new node $i \in N$ with $\text{ns}(i) = p$ **do** ▷ $\Theta(|p| \cdot |N|)$
    for each node $j \in N$ with $\text{ns}(j) \neq p$ **do**
      if $\cos(X_i, X_j) < \theta_{\text{block}}$ **then continue** ▷ blocking, eq. (6)
      end if
      compute σ_{ij}, ω_{ij} (eq. (7)), ρ_{ij}
      if (i, j) accepted by eq. (8) **then**
        $c_{ij} \leftarrow$ confidence (eq. (9))
        $\Lambda \leftarrow \Lambda \cup \{\, i \xrightarrow{\text{SAME\_AS}\{c_{ij}\}} j \,\}$
      end if
    end for
  end for
end for
return Λ

analyser parses heterogeneous timestamp strings, orders events chronologically, computes the overall time span, and flags inconsistencies—large gaps between consecutive events and contradictions between explicit PRECEDES/FOLLOWS edges and parsed timestamps. Such gaps and contradictions are exactly the cues an investigator looks for (an unexplained interval, a statement that is chronologically impossible).

Hypothesis support retrieves a case's connected evidence—persons, locations, patterns, mechanisms, and (for bloodstain experiments) per-stain statistical features—and asks an LLM to produce a primary hypothesis with explicit supporting and contradicting evidence, alternative hypotheses, and a reasoning chain. The hypothesis is stored as a Hypothesis node linked to the case and embedded, so it becomes part of the queryable, semantically searchable graph. We emphasise the framing: the system produces auditable hypothesis support, not a verdict. Every hypothesis points back to the specific evidence nodes behind it (fig. 5), so a human analyst can inspect, accept, or reject it on the basis of the cited evidence.

V. Implementation

The framework is realised as a prototype used solely to obtain the measurements reported here; none of the contribution depends on these choices. Extraction and reasoning run in Python; the typed graph and its cosine vector index are stored in Neo4j [4]; classical image features use OpenCV [27]; and extraction, embedding, and vision calls use a hosted LLM. Interface details are immaterial to the research questions and are omitted.

VI. Experimental Setup

A. Datasets

Table IV summarises the data. For text we use real Indian court judgments [28]; 28 well-documented real

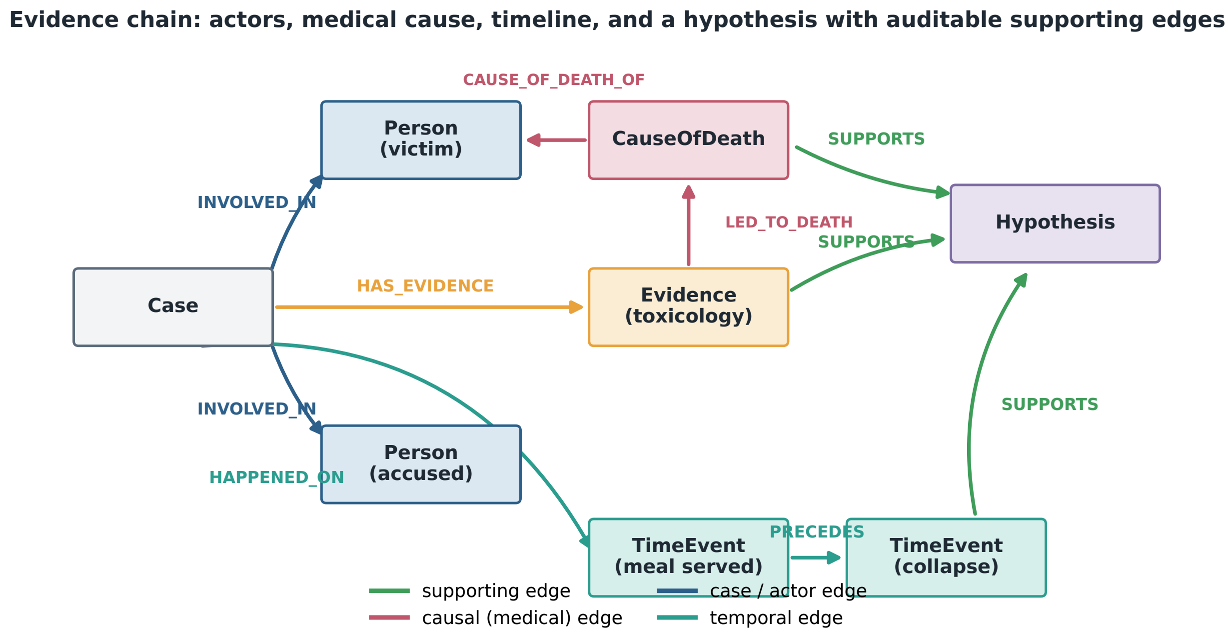

Fig. 5. An example evidence chain. A hypothesis is connected to the specific toxicology evidence, medical cause of death, and timeline events that support it; supporting and contradicting edges make the reasoning auditable.

TABLE IV
Datasets used for development and evaluation. Synthetic gold is useful for regression testing but circular for validity; documented real cases and physics-grounded image data provide the stronger external evidence.

Dataset	Source	Items	Role / modality	Ground-truth alignment
Indian court judgments	Public records / HF [28]	38	Real legal text extraction	Document text (no per-entity gold)
Documented real cases	Court records / public sources	28	Real-case entity validation	Two-annotator documented facts
Synthetic FIR / PM / lab / deposition / SOCO	LLM-generated	120	Text extraction regression	Self-consistent (circular)
Attinger bloodstain [13]	Published (2018)	61 exp.	Bloodstain image analysis	Physics-grounded mechanism
SOCOFing [29]	Kaggle	110,540	Fingerprint image analysis	Hand/finger label (not pattern class)
CEDAR signatures [30]	Public release	2,640	Document forgery detection	Genuine vs. forged
AZH wounds [31]	Public release	1,867	Wound image analysis	Clinical class (not forensic type)
Synthetic ballistics / tool marks	Procedurally generated	40	Image analysis regression	Known synthetic parameters

Indian criminal cases with human-curated factual gold (e.g. the Koodathayi cyanide killings and the 2024 R.G. Kar case), curated from authoritative public sources; and synthetic FIR, post-mortem, laboratory, deposition, and SOCO reports used for regression. For images we use the Attinger impact-spatter bloodstain dataset [13], SOCOFing fingerprints [29], CEDAR signatures [30], AZH wound images [31], and procedurally generated ballistics and tool-mark images.

B. Two evaluation tracks

We separate two tracks with deliberately different epistemic status. Track 1—synthetic-gold regression. The 120 synthetic FIR, post-mortem, laboratory, deposition, and SOCO documents carry LLM-generated gold. Because the gold and the system under test are the same class of model, agreement measures self-consistency, not correctness; we use it only to detect regressions and never report it as external evidence. Track 2—real-case validation (primary). The 28 well-documented cases carry gold curated from authoritative public records—documented facts not produced by any LLM. A first annotator curated the gold from the cited sources (dropping two unnatural-death cases as edge cases of non-canonical crime type, and normalising entries such as name spellings and the principal location); a second annotator then independently re-annotated all 28 cases blind to the first. Inter-annotator agreement is high—Cohen's $\kappa = 0.95$ on person roles and 1.0 on crime type (section VII-C, table VII)—which validates the first annotator's curation as the gold used here; the full disagreement list is released for adjudication. The pipeline extracts entities from a factual narrative and is scored against those facts. This is our primary result because it breaks the circularity of Track 1.

C. Matching and image metrics

Real-case matching is type-specific: persons, locations, and weapons use fuzzy string matching; crime type uses canonical-keyword matching against the case description (where the FIR extractor records the crime); and timeline events use date-aware matching on extracted year/month against documented dates, so descriptively named events are compared fairly to dated gold. Image analysis uses per-attribute accuracy for categorical attributes, precision/recall/F1 for binary attributes, and mean absolute error for numeric attributes, and is stratified by ground-truth alignment: aligned (bloodstain, physics-grounded), proxy (ballistics caliber), and misaligned (fingerprint hand/finger labels for a pattern-class task). A methodological safeguard is a label-free item identifier: the filename stem in several public datasets encodes the ground-truth label, so passing it as metadata would leak the answer into the prompt; removing this leak corrected an early, falsely perfect fingerprint result (section VII).

D. Configuration and reproducibility

Unless stated otherwise, extraction uses GPT-4.1 with strict JSON-schema structured outputs (falling back to JSON-object mode where the schema mode is unsupported) at decoding temperature 0.1; node embeddings are 1536-dimensional. We report the mean (and standard deviation) over three independent runs per condition, plus a case-level bootstrap 95% CI over the 28 cases (section VII); at this temperature all types are stable (F1 sd ≤ 0.03, and ≤ 0.01 for all but Location). Cross-case resolution uses an embedding blocking floor $\theta_{\text{block}} = 0.78$ and a name threshold $\theta_{\text{name}} = 0.88$ (eqs. (6) and (8)). The exact 28-case list, their public sources, gold composition, per-case scores, and the curator-plus-review curation protocol are given in appendix A; the documented-fact gold and the experiment scripts are released with the paper.

E. Controlled comparisons

We run two controlled ablations: ontology-guided extraction vs. a naive, same-LLM JSON baseline (section VII-G), which isolates the schema guidance of eqs. (1) and (2); and a CV-only vs. LLM-only vs. CV+LLM ablation of the bloodstain image pipeline (section VII). The remaining single-component comparisons (removing the graph; ungrounded querying, eq. (5); and disabling resolution, eq. (8)) are left to future work (section XI); we report measured ablations, not the full matrix, rather than approximate the rest with unmeasured numbers.

TABLE V
Research questions, supporting evidence, and main findings.

RQ	Evidence	Main finding
RQ1	ablation (fig. 8)	ontology guidance lifts macro-F1 0.53→0.73 and recovers Weapon 0.00→0.72
RQ2	section VI-B, table VI	synthetic gold is consistency-only; real-case micro-F1 0.78 (CI $[0.74, 0.82]$, n=28)
RQ3	table VIII	only bloodstain has task-aligned ground truth (F1 1.0, full set n=61)
RQ4	section VII-E, table X	precision-oriented SAME_AS: 27/28 accepts correct (1 common-name merge), recall 0.15; recovers the 3 genuine shared suspects

VII. Results

Table V maps each research question to its evidence and headline finding; the subsections that follow elaborate each in turn.

A. Synthetic regression (consistency check, not headline)

On the 120 synthetic documents the pipeline reproduces the LLM-generated gold with consistently high, stable agreement across runs, and every extracted structure satisfies the ontology (eqs. (1) and (2)). Per section VI-B, this is a regression signal only: it confirms stability and schema compliance but says nothing about real-world correctness, so we do not interpret it as evidence and report the real-case track below as our headline result.

B. Real-case validation (primary result)

Table VI and fig. 6 report the full system on the 28 documented cases (two-annotator gold, section VII-C; section VI), averaged over three independent runs. Person, crime-type, weapon/method, and dated-event extraction are strong and stable: F1 = 0.88 ± 0.01, 0.91 ± 0.01, 0.72 ± 0.01, and 0.88 ± 0.01 respectively, with standard deviations at or below 0.01. TimeEvent matching scores against the structured date field, where the pipeline records each event's calendar date in ISO form. The matching rule is deliberately explicit: a gold date counts as recovered when the extracted date shares its year, refined to year-and-month when both sides name a month; day-level exactness is not required. This year-or-year-month rule (rather than exact-ISO matching) matches the gold's own granularity—several documented dates are recorded year-only (e.g. a conviction in "2003")—and the identical rule scores both the full system and the baseline, so it cannot favour one. An earlier evaluation that read only the free-text event name (not the date field) under-counted recovery at F1 = 0.09, conflating “stored in a separate field” with “not extracted”—exactly the schema-compliance-versus-validity distinction this paper argues for (section VII-F). Location is the one genuinely weak type (F1 = 0.27) because the gold names a single principal location while the narrative mentions many places. Overall micro-F1 is 0.78 (±0.01) and macro-F1 0.73 (±0.01); the small gap between them reflects that only Location lags. We regard the strong, low-variance recovery of persons, crime types, weapons, and dates on real data—validated against documented facts no synthetic benchmark could provide—as the substantive result. Per case, full-system F1 ranges from 0.60 (Soumya, train) to 0.94 (Dabholkar), mean 0.78 (table XII). A case-level bootstrap (B=2000, resampling the 28 cases) places the 95% confidence interval for overall F1 at $[0.74, 0.82]$ (micro) and $[0.68, 0.78]$ (macro).

To check whether the strict per-mention metrics understate practical usefulness, we add two complementary measures. Principal-location recall—does any extracted location match the single gold principal location under the strict fuzzy matcher?—is 0.39 (11/28 cases); much of the gap to the precision-driven F1 (0.27) is additional, genuinely-mentioned locations counted as false positives, and many “misses” are more specific than the gold (section VII-F). The human inter-annotator study (section VII-C) shows this is granularity, not error: two annotators name the same place but at different specificity. Quantifying that, a granularity-tolerant principal-location recall—which credits the gold place when an extracted location names it at any granularity (token superset or subset)—reaches **0.86** (24/28). The cases it credits over the strict matcher are the same place named more or less specifically (“Vasant Vihar” vs. “Vasant Vihar, New Delhi”; “Koodathayi” vs. “Koodathayi, Kozhikode”); it deliberately does not credit genuine mismatches (Chhaparvad vs. Randhikpur), and even under-counts—missing cases where the system named only a sub-scene (“St Pius X Convent” for Kottayam). The system therefore recovers the correct principal place far more often (≈0.86) than the strict per-mention metric (0.27) suggests; what it lacks is principal-location selection, not place recognition. Date coverage—the fraction of documented years recovered anywhere in the extracted timeline—is 0.95, confirming that the pipeline does recover the documented dates: the looser coverage metric and the corrected per-mention TimeEvent score agree. The one genuinely weak type is Location: partly a matching-design issue and partly a principal-location selection problem (the system does not yet single out the principal scene from secondary places).

C. Inter-annotator agreement

To validate the documented-fact gold beyond a single curator, a second annotator independently re-annotated all 28 cases from the case narratives, blind to the first annotator’s gold; table VII reports agreement. On the two genuinely categorical decisions—each shared person’s role and each case’s primary crime type—Cohen’s κ is 0.95 and 1.0 (“almost perfect” and “perfect” on the Landis–Koch scale). On the open-set mention categories, where κ is degenerate (no closed true-negative universe) and pairwise F1 is the standard agreement metric, agreement is high for Person (0.86), CrimeType (0.97), TimeEvent (0.93), and Weapon (0.75). Location agreement is low (0.36)—and tellingly so: manual inspection shows the two annotators name the same place in 16 of 18 “disagreements,” differing only in granularity (“Munirka” vs. “a private bus in Munirka”; “Bagiya restaurant” vs. “Bagiya barbecue restaurant, Ashok Yatri Niwas hotel”). That two humans agree on the place yet score F1 = 0.36 is direct, human-grounded evidence that the system’s low Location score (section VII) is a metric/representation artefact, not an extraction failure. The few genuine disagreements are defensible edge cases (a witness who was also assaulted; a pathologist labelled officer vs. forensic_analyst; a driver-turned-approver labelled witness vs. accused); they are documented in the released agreement report rather than silently merged. The evaluation gold is the first annotator’s

TABLE VI
Real-case validation (full system) on 28 documented Indian criminal cases (two-annotator gold): mean over three independent runs (gpt-4.1, temperature 0.1); the F1 column reports mean ± standard deviation. Scores are against human-curated documented facts, not LLM-generated gold.

Entity type	P	R	F1 (mean±sd)
Person	0.82	0.95	0.88 ± 0.01
CrimeType	0.89	0.94	0.91 ± 0.01
Weapon / method	0.79	0.66	0.72 ± 0.01
Location	0.19	0.44	0.27 ± 0.03
TimeEvent	0.85	0.91	0.88 ± 0.01
Overall (micro)	0.72	0.86	0.78 ± 0.01
Overall (macro)	0.71	0.78	0.73 ± 0.01

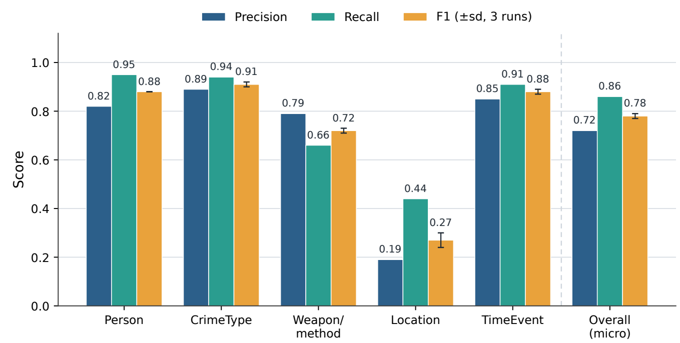

Fig. 6. Full-system precision, recall, and F1 by entity type on the 28 documented cases (mean over three runs; F1 error bars are ±1 sd). Persons, crime types, weapons, and dated events are recovered strongly and stably; Location is the one weak type.

TABLE VII
Inter-annotator agreement on the 28-case gold (two independent annotators). Cohen's κ for the categorical decisions; pairwise F1 (positive specific agreement) for the open-set mention categories, where κ does not apply.

Decision	Metric	Agreement
Person role	Cohen's κ	0.95
Primary crime type	Cohen's κ	1.00
Person mention	pairwise F1	0.86
CrimeType set	pairwise F1	0.97
TimeEvent dates	pairwise F1	0.93
Weapon	pairwise F1	0.75
Location	pairwise F1	0.36†

†Same place, different granularity in 16/18 cases (text).

curation, which this high agreement validates; because the system metric matches person names (not roles), the few role disagreements do not affect the reported scores. One caveat: the second annotator worked from the case narratives while the first curated from the cited sources, so the agreement also spans that source-to-narrative representation gap—which makes the reported κ a conservative estimate of curation reliability rather than an inflated one. A forensic-expert third pass (section XI) remains the natural next step.

D. Image analysis

Table VIII and fig. 7 report image evaluation, and they deliberately include negative results. Bloodstain impact-spatter recognition reaches F1 = 1.0 on the full Attinger impact-spatter set (n=61, every sample scored, none skipped)—the one modality with real, physics-grounded ground truth. We still read this conservatively: it is a coarse binary detection metric (the dataset is all impact-spatter, so there are no hard negatives) on a single dataset and the prompt was not separately held out. It is therefore evidence of full-set impact-spatter detection on one dataset, not general bloodstain-pattern recognition or a broad image-forensics claim. Wound detection (present vs. absent) over-detects (recall 1.0, precision 0.5). Ballistics caliber is recovered at roughly chance-level accuracy on synthetic data. The remaining tasks fail: single-image document forgery detection is near zero because deciding genuineness generally requires a reference exemplar; tool-mark tool-type classification is near zero on synthetic marks; and fingerprint pattern recognition is near zero because the dataset labels encode hand and finger position, not pattern class—the ground truth does not contain the attribute the task needs. We report these faithfully: they are not defects to be hidden but a map of the framework's operating envelope, and they expose a concrete evaluation-validity bug. The label-leak fix of section VI moved the reported fingerprint score from a false 100% (the filename label leaking into the prompt) to an honest $\approx 0\%$—a swing that is itself evidence for why honest, leakage-free evaluation matters.

TABLE VIII
Image evaluation, including honest negative results. "Aligned" ground truth encodes the attribute the task requires; "misaligned" does not. †Full Attinger set (n=61); single-dataset, coarse binary detection metric.

Image task	Metric	Score	Ground-truth status
Bloodstain (impact spatter)	F1	1.00†	physics-grounded, aligned
Wound (present/absent)	P/R	0.50 / 1.00	partial (over-detects)
Ballistics (caliber)	acc.	≈0.50	proxy / synthetic
Document (forgery)	F1	≈0	hard (needs exemplar)
Tool marks (tool type)	acc.	≈0	misaligned / synthetic
Fingerprint (pattern)	acc.	≈0	misaligned labels

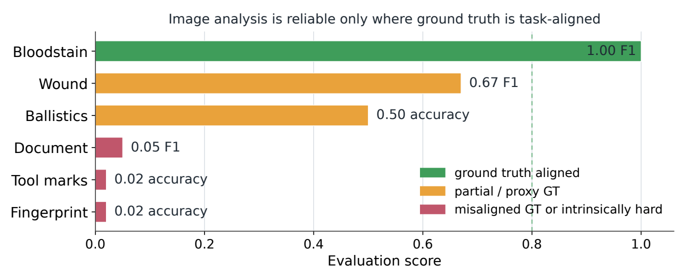

Fig. 7. Image analysis is reliable only where ground truth is task-aligned. Bloodstain (physics-grounded) succeeds; tasks with misaligned or proxy ground truth, or that are intrinsically hard from a single image, do not.

Hybrid ablation: On the full impact-spatter set (n=61) we ablate the image pipeline into CV-only, LLM-only, and the full CV+LLM hybrid (table IX). Classical CV features alone produce no pattern label, so CV-only detection is F1 = 0—the CV stage extracts stain geometry, not a semantic class. The LLM vision step alone detects impact spatter on all 61 samples (F1 = 1.0), and the full hybrid matches it (1.0). The honest reading is that the LLM drives the high-level classification and the CV stage is necessary only for the quantitative per-stain features it adds to the graph; on this coarse binary metric the hybrid gains no accuracy over LLM-only. This tempers, rather than supports, a strong hybrid claim on this task.

E. Cross-case resolution (RQ4)

We evaluate resolution over all 28 cases and, to expose recall, label every candidate pair, not only the accepted ones (table X). In a single run, the 239 resolvable entities admit 9973 cross-case pairs; embedding blocking keeps 243; the conservative decision (eq. (8)) accepts 28. Precision is 27/28 = 0.96: the accepts comprise 16 crime-type category links, 7 recurring person/organisation identities, 3 generic weapon-type links, and 1 location. Crucially,

TABLE IX
Bloodstain image ablation (impact-spatter detection, full set n=61). CV alone yields no semantic label; the LLM drives detection; the hybrid adds no accuracy over LLM-only on this coarse metric.

Configuration	F1
CV-only (no LLM)	0.00
LLM-only (no CV)	1.00
CV + LLM (hybrid)	1.00

TABLE X
Cross-case SAME_AS resolution over all 28 cases (one extraction run). Every candidate pair was labelled by an objective cross-case identity rule.

Stage	Pairs
All cross-case pairs (same type)	9973
After embedding blocking (candidate pool)	243
Accepted SAME_AS links	28
true accepts (TP)	27
false accepts / merges (FP)	1
True identities missed in pool (FN)	151
Candidate-set precision 0.96, recall 0.15, F1 0.26	

the resolver recovers the three genuine shared-suspect links the dataset contains by construction—the gang members Ravi Kapoor, Amit Shukla, and Baljeet Malik, shared by the Jigisha Ghosh and Soumya Vishwanathan murders (confidence 0.80, name-plus-context), plus the recurring Central Bureau of Investigation (three cases) and Delhi Police. The single false merge is instructive: a witness "Ajay Kumar" in the Nitish Katara case is linked to an accused of the same common name in Soumya Vishwanathan, accepted on the identical name alone (confidence 0.60, the lowest of any accept) despite the role disagreement—the bare-common-name collision the design tries to suppress. Candidate-set recall is 0.15 (27 of 178 pool identities): the 151 misses are overwhelmingly identical crime types (146, Murder = Murder across cases) the decision rejects for lacking corroborating context—a deliberate choice not to link on a bare common word. Candidate-set F1 is thus 0.26, quantifying the high-precision / low-recall trade-off the design intends over a 5.4× larger search space than the ten-case pilot—a candidate-set, not a global, benchmark (should-link labels follow the objective rule of table X). Links are non-destructive, so a cross-case query traverses SAME_AS edges without collapsing provenance.

F. Error analysis

Table XI summarises the dominant, recurring error patterns; we report patterns rather than cherry-picked instances. The dominant remaining error is a single representation/matching mismatch rather than a perception failure: Location over-listing. The gold lists one principal location per case while the extractor faithfully returns

TABLE XI
Representative error patterns by entity type (real-case set).

Type	Dominant error	Likely cause
Location	over-listing; low precision	gold has 1 principal place; model lists all mentions
TimeEvent	apparent low precision (metric only; resolved)	date stored in structured date field; name-only scoring under-counted
Person	occasional FN on minor actors	person named once in passing
Weapon / method	rare FN	method stated narratively, not as a noun

every place mentioned, so precision is 0.19 though the principal location is recovered in 11/28 cases. It is fixable by binding the case to its principal event/location (via HAPPENED_ON/OCCURRED_AT) and scoring against that. Concretely, the Location "false positives" are real, relevant places that are often more specific than the gold: for Abhaya the system extracts "St Pius X Convent" (the actual scene) against the gold's "Kottayam"; for Dabholkar, "Omkareshwar bridge" against "Pune"; and for Gauri Lankesh, "the driveway of her home" against "Rajarajeshwari Nagar". These are not wrong—underscoring that for Location the metric, not the extraction, is the limiting factor.

TimeEvent illustrates the same principle at the level of where a value is stored. The two-stage extractor records each event's calendar date in a structured date field (e.g. 2017-09-05 for Gauri Lankesh) and gives the node a descriptive name ("Murder of Gauri Lankesh"); an evaluation that read only the name scored TimeEvent at F1 = 0.09 and date coverage at 0.10, suggesting a recovery gap. Reading the field the schema designates for dates—as the system itself does when querying—raises TimeEvent to 0.88 and date coverage to 0.95. The dates were always extracted; the earlier metric conflated schema location with non-recovery, precisely the validity confusion this paper warns against. (The naive baseline, instructed to embed the date in the event name, scores 0.91 either way.)

G. Ablation: the value of ontology guidance

To isolate the contribution of ontology-guided extraction we compare the full system against a baseline that uses the same LLM and JSON output but a single generic entity-extraction prompt—no forensic ontology type descriptions, no per-type rules, and no two-stage relation grounding—scored by the identical matcher over the same 28 cases, three runs each (fig. 8). The result is decisive. Ontology guidance is what makes the structured forensic types recoverable at all: it lifts Weapon/method from F1 = 0.00 to 0.72—the baseline records no usable weapon across any case (it never tags, e.g., cyanide, the

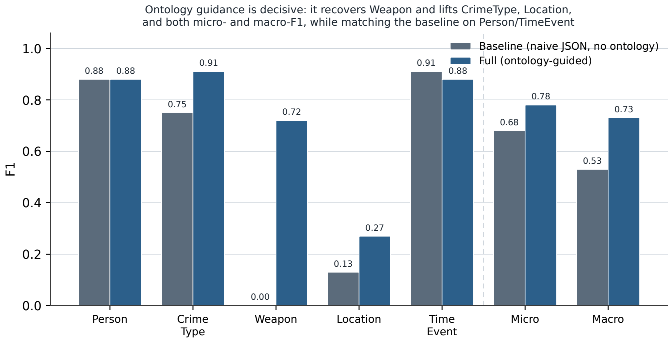

Fig. 8. Ablation: ontology-guided full system vs. a naive JSON baseline (same LLM, no ontology), F1 by type and overall, mean over three runs. Ontology guidance is decisive for Weapon, CrimeType, and Location and wins on both micro- and macro-F1, while matching the baseline on Person and TimeEvent.

Koodathayi murder agent, as a Weapon/method), whereas the ontology-guided extractor recovers weapons at recall 0.66—lifts CrimeType from 0.75 to 0.91 and Location from 0.13 to 0.27, and matches the baseline on Person (0.88) and TimeEvent (0.88 vs. 0.91). The full system therefore wins on both aggregate measures: macro-F1 0.73 vs. 0.53 and micro-F1 0.78 vs. 0.68. (An earlier evaluation that scored TimeEvent from the event name alone—see section VII-F—made the baseline appear to win micro-F1; reading the structured date field removes that artefact.) The honest reading is that ontology guidance helps wherever forensic typing matters and costs nothing on the types both systems already handle. A further design choice with a measured effect reported above is the label-free evaluation fix that corrected a false fingerprint score from 100% to $\approx$ 0% (section VI). A second ablation isolates the bloodstain image pipeline (CV-only vs. LLM-only vs. CV+LLM), reported in section VII. Remaining comparisons—removing the graph, ungrounding the query generator, and disabling resolution—are left to future work (section XI).

VIII. Discussion

Why graph structure helps: Representing a case as a typed graph makes the relationships that matter to investigation—who was involved, what caused death, what preceded what, which evidence supports which hypothesis—explicit and traversable. Multi-hop and cross-case questions become graph patterns rather than brittle joins, and evidence chains become first-class objects that a human can audit.

Why real-case validation matters: Validating against documented facts of real cases is the difference between measuring self-consistency and measuring correctness. Our most defensible findings—stable recovery of persons, weapons, and crime types (F1 $\approx$ 0.86–0.88, sd $\leq$ 0.03) and the measured ablation gain from ontology guidance—come from this protocol precisely because the gold was not produced by an LLM.

Why the location score is low: Location is low for a structural reason: the gold records the principal location while the extractor returns every place mentioned, so genuinely-relevant secondary places count as false positives. The fix is structural—bind the case to its principal location (via OCCURRED_AT) and score against that—a concrete, near-term improvement rather than a fundamental limitation. It shows that part of the value gap lies in evaluation design, not only in extraction—the same lesson the TimeEvent date-field correction (section VII-F) makes plain.

Why image results contain negatives: The image negatives are informative. They separate genuine model limitations (single-image forgery detection is hard without an exemplar) from evaluation artefacts (fingerprint "failure" is really a ground-truth mismatch). Reporting both, instead of selecting the one task that works, is what makes the bloodstain success credible.

Why non-destructive resolution is safer: In forensics a fabricated connection can mislead an investigation. Linking rather than merging means the system can surface a possible cross-case connection while preserving the ability to reject it, and it never destroys the source provenance that an analyst needs to evaluate the link.

LLM hallucination and its mitigation: The framework mitigates hallucination through four reinforcing mechanisms: ontology validation (typed filtering of outputs), strict structured output (schema-constrained generation), retrieval grounding (querying anchored in real labels and paths), and evidence-chain reporting (every hypothesis cites its supporting nodes). None eliminates hallucination, but together they bound it and make residual errors visible and auditable.

IX. Threats to Validity

Internal validity: Extraction and scoring share fuzzy-matching heuristics, so a lenient matcher could over-credit extraction. We mitigate this with exact documented-fact gold, conservative matchers, and per-type reporting that would expose leniency—location precision is reported low (0.19), not masked. LLM stochasticity is a second internal threat: schema constraints bound, but do not remove, run-to-run variance; we report mean $\pm$ standard deviation over three runs and a case-level bootstrap 95% CI ($[0.74, 0.82]$ for overall micro-F1). The remaining gaps are listed as future work: the full baseline matrix beyond the ontology ablation (section VII-G) and a still larger, more routine case set. With 28 cases a formal significance claim would still be premature, so we present the numbers as indicative.

External validity: The 28 real cases (and small per-modality image samples) skew toward high-profile Indian cases; the results may not transfer to routine casework, other jurisdictions, or other languages. Synthetic text is deliberately excluded from our headline claims for exactly this reason (section VI-B).

Construct validity: Our metrics measure recovery of entities and relations against documented facts—a proxy for, not a guarantee of, investigative usefulness. The image negatives are partly a construct problem: a dataset label (finger position) can fail to operationalise the intended construct (pattern class), so a low score reflects measurement mismatch rather than model incapability. Where the construct is well operationalised (bloodstain mechanism), the same pipeline scores at ceiling.

Ethical and legal validity: The framework produces hypothesis support, not determinations. Any deployment raises privacy, bias, security, and chain-of-custody concerns and would require expert oversight and governance. We make no claim of forensic validity or legal admissibility, and no output should serve as the sole basis for any investigative or legal decision.

X. Limitations

We state limitations explicitly. The system should be treated as an exploratory research prototype: no claim is made that extracted graph facts, SAME_AS links, or generated hypotheses meet forensic admissibility standards. (1) The system is decision and hypothesis support, not a replacement for forensic experts, and we make no claim of forensic correctness beyond the measured evidence, nor any claim of legal admissibility. (2) Extraction and reasoning depend on hosted LLMs, which raises reproducibility concerns: outputs can vary across model versions and sampling. (3) Access to real forensic data is limited; much of our text data is synthetic and therefore weak for external validity, which is why we foreground the real-case protocol. (4) Image tasks require domain-aligned ground truth; several public datasets do not encode the attribute the forensic task needs. (5) Single-image forensic interpretation is intrinsically limited for comparison tasks (forgery, tool-mark, ballistic matching) that need reference exemplars or multi-view evidence. (6) Real-world deployment would require audit logging, security, privacy protection, and expert review that are out of scope here. (7) Evaluation sample sizes—28 real cases and small per-modality image samples—remain limited, so the reported numbers should be read as indicative rather than definitive.

XI. Future Work

Several directions follow directly. Powered, baselined evaluation is the immediate priority: completing the controlled component matrix beyond the two ablations reported here (removing the graph, ungrounding the query generator, and disabling resolution); and expanding real-case validation beyond 28 cases with formal significance testing. Expert annotation: extend the two-annotator gold (section VII-C) with forensic-expert annotators to enable rigorous, attribute-level validation. More real data: incorporate larger corpora of real FIR, police, and laboratory documents, and multilingual Indian legal and forensic text. Multi-view image evidence: support exemplar-based and multi-view comparison for forgery, tool-mark, and ballistic tasks that single images cannot resolve. Domain-specific models: distil smaller, cheaper extraction models specialised to forensic schemas to improve reproducibility and cost. Human-in-the-loop: add annotation and active learning so analyst corrections improve extraction over time. Uncertainty calibration: attach calibrated confidence to extracted facts and hypotheses. Provenance and integrity: add chain-of-custody-aware, versioned graph provenance so that every node and edge carries an immutable, auditable history.

XII. Conclusion

We presented an ontology-guided, multimodal forensic knowledge graph framework that unifies heterogeneous legal, textual, and visual evidence into one type-validated, embedded, queryable graph, with a reasoning layer for semantic retrieval, grounded querying, temporal reconstruction, conservative cross-case entity resolution, and auditable hypothesis support. On the documented facts of 28 real Indian criminal cases (two-annotator gold, three runs each) the framework recovers persons, crime types, weapons, and dated events strongly and stably (F1 = 0.88, 0.91, 0.72, 0.88; sd $\leq$ 0.01), with overall micro-/macro-F1 = 0.78/0.73 (bootstrap CI $[0.74, 0.82]$); an ablation shows ontology guidance is what makes the structured forensic types recoverable and lets the full system lead on both micro- and macro-F1, with Location the one remaining weak type. Image analysis is reliable only where ground truth is task-aligned; we report the remaining near-zero scores as evidence of evaluation honesty. We do not claim to automate forensic judgement: the framework structures evidence and supports hypotheses for a human expert, and establishing forensic validity would require expert-annotated data and prospective study, which we leave to future work. Within those bounds, the contribution is a reproducible framework and an evaluation methodology—in particular a documented-fact protocol—that can help discourage circular benchmarking in forensic AI.

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Appendix

A. Real-case list and gold-curation protocol

Table XII lists the 28 documented cases and the composition of their gold (counts of gold Person/Location/Weapon/TimeEvent entries). Gold annotations were curated from authoritative public records—the English-language encyclopedic summary of each case and contemporaneous court reporting—by a first annotator, who dropped two unnatural-death cases (the Burari deaths and the Sunanda Pushkar death) whose "crime type" does not map to a canonical category and normalised entries (name spellings, the choice of principal location, weapon/method). A second annotator then independently re-annotated all 28 cases from the narratives, blind to the first annotator's labels; the two annotations agree at Cohen's $\kappa = 0.95$ on roles and 1.0 on crime type (section VII-C, table VII), validating the first annotator's curation as the gold used here (the full disagreement list is released). For each case we recorded the named persons and their roles, the single principal location, the weapon or method, the crime type(s), and the documented dates/time-events, recording only facts asserted by the sources and not inferences. The pipeline is run on a short factual narrative of each case (distinct from the gold) and scored against these facts. The bias toward high-profile, well-documented cases remains a limitation (section IX); a still broader, less self-selected case set would further strengthen external validity.

B. Ontology and relationship summary

Table I lists node groups; the 48 relationship types fall into involvement, spatial/temporal, causal/medical, evi-

TABLE XII
The 28 documented real cases (two-annotator gold), gold composition (P/L/W/T = gold Person/Location/Weapon/TimeEvent counts), and per-case full-system F1 (mean over three runs), sorted by F1. Sources are the corresponding English Wikipedia articles and contemporaneous court reporting.

Case	P	L	W	T	F1	Case	P	L	W	T	F1
Murder of Narendra Dabholkar	3	1	1	2	0.94	Murder of Soumya Vishwanathan	6	1	1	2	0.78
Murder of Gauri Lankesh	3	1	1	1	0.88	Swathi murder case	2	1	1	3	0.78
Koodathayi cyanide killings	10	1	1	6	0.88	2008 Aarushi–Hemraj murder	4	1	2	4	0.77
Assassination of Phoolan Devi	2	1	1	2	0.88	Kathua rape & murder	4	1	2	3	0.77
Murder of Jigisha Ghosh	4	1	0	3	0.87	Murder of Pradyuman Thakur	2	1	1	1	0.75
Murder of Priyadarshini Mattoo	2	1	2	3	0.87	Nithari serial murders	3	1	1	4	0.73
2012 Delhi gang rape & murder	8	1	1	3	0.87	Saravana Bhavan (Santhakumar)	3	1	0	3	0.73
Tandoor murder (Naina Sahni)	4	1	1	4	0.83	2020 Hathras gang rape & murder	4	1	0	3	0.72
Murder of Jessica Lal	4	1	1	3	0.82	Murder of Graham Staines	5	1	1	2	0.70
Murder of Neeraj Grover	3	1	1	2	0.82	Bilkis Bano case	2	1	3	3	0.67
Murder of Pramod Mahajan	2	1	1	3	0.82	2024 R. G. Kar rape & murder	2	1	1	3	0.67
Sister Abhaya murder	3	1	1	2	0.78	Murder of Shraddha Walkar	2	1	1	2	0.67
1999 Delhi BMW hit-and-run	6	1	1	2	0.78	Nitish Katara murder	8	1	1	3	0.61
Murder of Sheena Bora	5	1	1	2	0.78	Soumya murder case (train)	2	1	0	2	0.60

Totals: 108 Person, 28 Location, 29 Weapon, 76 TimeEvent gold entries across 28 cases; mean per-case F1 = 0.78 (range 0.60–0.94).

dential, image-derived, and cross-case families. Controlled enumerations (e.g. PatternType, InjuryType, FingerprintClass, ToolMarkType) constrain selected attributes at insertion.

C. Baseline extractor (ablation)

The naive baseline of section VII-G uses the same model and JSON output as the full system but a single generic prompt with no ontology type descriptions, no per-type rules, and no two-stage relation pass. It asks the model to extract entities of type Person, Location, Weapon, CrimeType, TimeEvent, or Case from the factual narrative into JSON ({entity_type, properties}), with Person carrying name and role and the others a name/type—and nothing about the forensic ontology, relationships, or evidence spans. Both conditions are scored by the identical matcher; the released experiment script contains the exact prompt.

D. Structured extraction schema (illustrative)

Entity extraction emits objects of the form {entity_type, entity_id, properties} constrained to a per-document JSON schema; relationship extraction emits {source_entity_id, target_entity_id, relationship_type, confidence, evidence_span}, where evidence_span is the exact supporting text. Identifiers are namespaced per document to prevent cross-document collisions.

E. Example Cypher query

A grounded NL-to-Cypher generation for "which cases share a cause of death linked to the victim?" yields a pattern such as:

```
MATCH (c:Case)<-[:INVOLVED_IN]-(p:Person)
  <-[:CAUSE_OF_DEATH_OF]-(cod:CauseOfDeath)
RETURN c.title, p.name, cod.primary_cause LIMIT 50
```

F. Ethical considerations

This framework is intended for research and as decision/hypothesis support under expert oversight. It must not be used as a sole basis for any legal or investigative determination. Deployment on real case data would require informed governance, access control, privacy safeguards, bias assessment, and human expert review. Reported metrics describe performance on the stated datasets only and do not constitute a claim of forensic validity or legal admissibility.